

Ethical Considerations in Governmental Cryptography

Since states first started encrypting messages in ancient times, governments have sought to control cryptographic techniques. These attempts to strengthen their own cryptographic technology to better protect sensitive communication while attacking adversaries' technologies to anticipate and counter their plans introduce a wide range of ethical questions concerning the right to privacy, the value of centralization, and the cost of security. This paper introduces the ways in which governments see cryptographic systems as both an asset and a barrier to achieving their goals, and discusses moral, legal, and scientific arguments over the proper role for governments in constructing and regulating cryptography.

It begins by providing an overview of how governments have embraced, undermined, and regulated cryptography, both in recent times and throughout history, including the roots of governments' interest in cryptography, governments' efforts to protect the privacy of their citizens' communications, and the ways in which governments have fought against strong cryptographic standards in the name of national security. It then considers three specific ethical issues, using case studies of specific policies and examples, while also considering perspectives from academic and technical literature.

The first concerns use of "backdoors" to provide law enforcement access to encryption systems, analyzing three instances of mandated backdoors and the impact those had on the security and reliability of cryptographic systems. The second considers individual privacy through the lens of the cypherpunk movement, exploring the ways in which citizens have sought to protect their communication from policies providing national security agents the authority to access interpersonal conversations. The final section examines freedom of expression, specifically interrogating whether governments deserve the right to define what content is considered safe and proper and presenting multiple, competing perspectives on what speech deserves protection.

1. How Governments Use Cryptography: History and Policies

Governmental interest in cryptography dates back to ancient Rome, birthplace of the Caesar cipher. Perhaps the most well-known encryption scheme in popular culture, the Caesar cipher encrypted messages by replacing each letter in the text with the letter that came three places later in the alphabet [1]; the letter A would be replaced with D, B with E, and so on. To decrypt a ciphertext, the recipient would shift each letter back three spots in the alphabet, undoing the original encoding.

The Romans' interest in encrypting their messages arose from the vast geographical extent of their empire, where communications had to travel long distances through potentially adversarial territory to get from one official to another. Maintaining secrecy of communications was particularly important given their content, which frequently included plans for battle and further expansion of great interest to opponents.

In the centuries after Caesar's invention, government investment into cryptography came primarily during times of warfare. Similar to the Roman empire, times of war saw the need to send messages of great importance across contested land where messengers were at high risk

of capture, creating a clear need to encode messages in case they were intercepted by opposing forces.

Nearly every American conflict gave rise to cryptographic innovations. Cipher methods in the American Revolution ranged from simple shift ciphers akin to the Caesar cipher to physical systems using paper cut-outs [1]. Officers would share templates that they used to cover their paper when writing and reading letters, putting the true message in the holes left uncovered by the sheet and filling the rest of the page with a fake message.

The invention of the telegraph at the dawn of the Civil War led to the development of new encryption methods that scrambled and jumbled words instead of encoding them with shifts and codebooks, a change necessitated by the increased volume of communication that the telegraph enabled [1].

The two World Wars brought even greater acceleration in cryptographic technology, with the American government inventing the first widely-used machines designed solely for encryption [2, 3]. In 1922, the United States Army invented the M-94 cipher cylinder, a device with 25 rotating disks with letters A-Z that could be used to scramble a message with a new daily key [1]. While invented by the military, similar technology was adopted by the U.S. State Department to protect their communications a decade later.

Simultaneously, the first commercial cryptography companies were being started in both the United States and in Europe, inventing devices like the electrical rotor, which automatically applied successive Vigenère ciphers on messages. These companies sold their products primarily to their nation's army and navy, including one company selling the German Navy the famous Enigma encryption system [1].

As technology progressed and the world grew increasingly interconnected, however, governments became interested in cryptography for national security outside of "traditional" warfare. After the end of the second World War and into the Cold War, the United States held the lead in cryptographic technology internationally, allowing the government to control access of the best systems from adversaries and to communicate securely while eavesdropping on foes [4].

As technology advanced, the United States government went to great lengths to protect that advantage. In 1992, fearing that advancement in cryptographic methods would jeopardize the ability of national security agencies and the military to decode others' communication, Congress enacted an export-grade cryptographic regime, which mandated that all cryptographic technology past a certain strength could only be sold in the United States, while weaker technology could still be sold abroad [4].

The creation of such an export regime, analogous to the restrictions placed on weapons and other dual-use technology with potential military applications, made clear the lens through which the United States saw cryptography: it was a strategic technology to control and keep out of the hands of international competitors.

The 1992 regime was hardly successful. Challenges in monitoring and preventing exports of software and the difficulty for technology providers to implement two different encryption protocols in their products resulted in weaker technology and poor enforcement. As a result, the export restrictions were largely repealed just a few years later, in 1996. The effect of the restrictions is still felt to this day, however, as many software products were updated to be

backwards compatible with the systems used in the early 1990s, leaving the flawed encryption protocols that were mandated for international sale in place and vulnerable to exploitation [4]. The export regime was just the beginning of an ever-present push from the government for access into cryptographic systems. Despite its failure, many in the government, especially in national security agencies, began pushing for “golden keys” or “backdoors,” special private keys that would give government officials access to all users’ communications with a certain technology or application [4]. This accelerated after September 11th attacks and ensuing “war on terror,” which saw far greater deference to national security and law enforcement agencies in bills like the PATRIOT Act. At the same time, this expansion in the power of law enforcement has been met with resistance by both citizens and private companies seeking to protect their clients’ privacy. This conflict came to a head in public light after terrorist attacks in San Bernardino, California in 2015, when Apple refused to remove the limit on password attempts on a suspect’s phone despite the FBI’s requests [5].

This pressure campaign for access into protected systems demonstrates the dual, competing ideals governments strive to balance when using and regulating cryptographic systems. On the one hand, cryptography is an asset for governments to protect their own communication as well as the privacy and freedom of their citizens. On the other, cryptography is a barrier to monitoring communications of potential criminals, foreign adversaries, terrorists, and other adversaries [6]. Striking the balance between these two competing aims is the primary ethical consideration for governments regulating and implementing cryptographic systems.

When viewing cryptography as a barrier, governments request backdoors and attempt to crack encryption of commonly used devices and applications. This raises significant ethical questions concerning the right to privacy and freedom from government interference in personal speech and communications. These concerns rose to the public sphere after incidents like Edward Snowden’s leaks revealing the National Security Agency had discovered and was exploiting flaws in encryption protocols used by technology giants like Google to collect and read communication from their users [7, 8]. It also raises concerns about the efficacy of such tactics, as leading researchers have argued that backdoors are impossible to implement securely [9]. These concerns were exacerbated by recent hacks by the Chinese government that exploited backdoors mandated by the 1994 Communications Assistance for Law Enforcement Act, which required telecommunications companies redesign their networks to enable law enforcement agencies to access customers’ communications [10]. These and other such attacks demonstrate how, even with the best of intentions, backdoors can be compromised to decrease security, further complicating the case for their implementation.

When viewing cryptography as an asset, governments adopt nationwide protocols for encryption and defend the freedom of their citizens to express themselves online and communicate securely. In 2021, the United Nations Human Rights Council adopted a resolution to protect human rights on the internet, including a recognition that encryption was “important to ensure the enjoyment of all human rights” on a 43-0 vote with 4 abstentions [11]. Individual countries have adopted their own encryption standards, like the United States’ Advanced Encryption Standard, a symmetric block cipher used for secure encryption since 2001. The United States has both used AES to encrypt unclassified government communications and encouraged private entities to use it for their own privacy and security [12].

Other than the aforementioned limits on exports, the United States has no regulations limiting the creation or use of encryption, allowing its development unfettered in the private sector [3]. Many other countries have taken a similar approach, including the European Union, whose only regulations on cryptographic technology concern exports to countries outside of the EU. Some other countries have opted for more aggressive regulation, prioritizing national security concerns over individual autonomy and privacy. China is one such example, with strict standards governing cryptographic technology that require products garner approval from the State Commission for Encryption Management before being sold or used.

Both the history of governmental utilization of cryptography and the policies governments currently apply to cryptography demonstrate the ethical dilemma encryption creates. The balance between individual privacy and national security is full of contention, with different situations emphasizing different aspects of the moral debate. The following sections explore both sides of the debate by narrowing in on three case studies: government-mandated backdoors, the cyberpunk movement, and freedom of expression.

2. Government-mandated backdoors

Encryption systems form the backbone of modern digital communication, protecting sensitive data and ensuring privacy and security. Backdoors are intentional vulnerabilities placed within cryptosystems made to enable access for authorized users. The government has a long track record of trying to install backdoors into many popular cryptographic systems. They argue that backdoors empower law enforcement to combat crime and terrorism, while critics contend that they inherently weaken systems, eroding trust and risking exploitation [13]. This tension between national security and individual privacy highlights one of the most contentious ethical dilemmas in modern cybersecurity.

To explore the numerous implications of backdoors, we will analyze three pivotal case studies. The first is the Dual EC DRBG algorithm [14], a cryptographic standard promoted by the NSA which contained an undisclosed backdoor, revealing the consequences of eroding trust in public standards. The second is the Clipper Chip initiative [15], a government effort to implement a key escrow system, which backfired due to economic and transparency concerns. Lastly, we examine the CALEA act [16], whose mandated wiretapping systems were recently hacked, underscoring the inherent insecurity of mandated access systems. These three case studies collectively illustrate the risks of embedding cryptographic backdoors and offer a lens through which to analyze their technical, ethical, and societal impact.

The Dual EC DRBG Backdoor

Random number generation is a crucial component of many cryptosystems, integral to generating everything from nonces to encryption keys to prime numbers [17]. Often, breaking the random number generator can lead to breaking the entire security system. The importance of this cannot be overstated in an era where robust encryption underpins modern digital security.

In 2006, the Dual Elliptic Curve Deterministic Random Bit Generator (Dual EC DRBG) algorithm was officially standardized as one of four cryptographically secure pseudorandom number generators (CSPRNGs) in the National Institute of Standards and Technology's (NIST) Special Publication 800-90A [14]. This endorsement elevated Dual EC DRBG to a position of

prominence, nearly mandating its use in critical systems and protocols. One notable example was its application in SSL/TLS encryption, a protocol widely used to secure communications over the internet. SSL/TLS underpins HTTPS, the mechanism that encrypts data transfers between users and websites. As a result, any vulnerabilities in Dual EC DRBG had the potential to expose massive amounts of sensitive data, impacting everything from email and instant messaging to voice over IP communications.

Dual EC DRBG is a cryptographically secure pseudorandom number generator based on the presumed difficulty of solving the Elliptic Curve Discrete Logarithm Problem (ECDLP). The algorithm operates by selecting two points, P and Q , on an elliptic curve E defined over a finite field $\mathbb{F}_p$. P is publicly known, while Q is known only to the algorithm's creator. The security of the generator relies on the assumption that Q is chosen independently and without any exploitable relationship to P .

At each iteration, Dual EC DRBG computes the pseudorandom output by deriving the x-coordinate of the elliptic curve point resulting from a scalar multiplication operation. Specifically, given the current secret seed t , the generator calculates tP to produce an intermediate point from which the x-coordinate is extracted. This x-coordinate is then truncated to generate a new seed s . We then calculate sQ and extract the x-coordinate. We truncate the result and return it as the pseudorandom output r . The new seed value s is then used as the starting t for the next iteration.

The primary vulnerability in Dual EC DRBG arises if Q is constructed with knowledge of a discrete logarithmic relationship to P [18]. Let d denote the private scalar such that $Q = dP$. If this relationship exists and d is known, an adversary observing sufficient output can reconstruct the generator's internal state by solving the truncated x-coordinate inversion problem. Once the state is recovered, the adversary can determine all subsequent outputs and, crucially, any previously generated outputs.

This exploit is exacerbated by the fact that the output truncation is minimal; enough information is retained for feasible recovery of the internal state. Additionally, P and Q are standardized and fixed, raising suspicion that Q was selected with a deliberate relationship to P . These conditions allow an entity with knowledge of d to entirely subvert the cryptographic security of any system employing Dual EC DRBG.

In 2013, the true scope of the issue became clear. Reports from Reuters [19] revealed that, prior to standardization, the NSA had secretly paid RSA Security \$10 million to make Dual EC DRBG the default in their BSAFE cryptographic library. Since BSAFE was widely used by developers as a foundation for secure applications, this deal ensured Dual EC DRBG's adoption across numerous critical systems. Leaked documents from Edward Snowden provided further evidence: the NSA had worked to become the sole editor of Dual EC DRBG during its standardization process and ensured that the algorithm contained a deliberate backdoor. By controlling the specification of Q , the NSA could use this backdoor to decrypt communications encrypted using Dual EC DRBG-based systems.

This controversy highlights the ethical breach inherent in prioritizing state surveillance over the collective trust that underpins global cryptographic systems. By embedding a backdoor into the algorithm, the NSA subverted the neutrality of the standardization process, exploiting NIST's role as an "honest broker" to institutionalize a compromised standard. This betrayal of trust had tangible consequences: Dual EC DRBG's adoption in widely used libraries such as

RSA's BSAFE resulted in insecure implementations across a wide array of critical systems. These vulnerabilities exposed sensitive information to exploitation, undermining the very security cryptographic systems are meant to provide. The algorithm's inefficiency – orders of magnitude slower than its counterparts – needlessly degraded the performance of technologies that relied on it, hindering innovation while offering weaker security.

NIST's close relationship with the NSA, codified through decades of collaboration under agreements like the 1989 Memorandum of Understanding, facilitated undue influence over the standardization process [20]. This historical entanglement enabled the NSA to suppress dissenting voices and bypass rigorous scrutiny. The resulting damage to NIST's credibility extended beyond technical circles; international partners and industries reliant on US-developed standards began to lose confidence, prompting efforts to explore alternative cryptographic ecosystems. Foreign governments questioned the impartiality of the US-endorsed standards. For example, European entities began investing in independent standardization processes to reduce reliance on US-dominated cryptography. This fragmentation of the global cryptographic community jeopardizes international cooperation on cybersecurity, emphasizing the need for transparent and inclusive processes that resist domination by any single actor [21]. To repair trust, ethical commitments to neutrality and transparency in the standardization process are essential.

Dual EC DRBG highlights a scenario in which the government intentionally obfuscated the existence of a backdoor. In the next case study, we will analyze a situation where the government embraced transparency by making the backdoor known.

The Clipper Chip Controversy

In the early 1990s, the US government sought a solution to ensure law enforcement could maintain its ability to intercept and decipher communications. The Clipper Chip, proposed under the Clinton Administration, was a controversial step toward embedding a surveillance mechanism directly into encryption standards [15]. This chip – to be used in telecommunications devices – implemented a "key escrow" system where encrypted communications could be decrypted using government-held keys, available only under court order [22]. Proponents emphasize that this initiative preserves privacy while equipping law enforcement to combat crime and terrorism in an increasingly digital age. However, the proposal's underlying assumptions and technical design attracted significant criticism, mirroring similar themes from the Dual EC DRBG controversy.

Proponents largely focus on the national security imperative. The administration argued that criminals, including terrorists, would need to use Clipper-enabled devices for legitimate communications such as with banks or suppliers, enabling the government to monitor key networks while maintaining oversight through the judicial system [15]. Numerous legal reviews supported these assertions, arguing that the system's safeguards aligned with constitutional protections under the Fourth Amendment and statutory frameworks like Title III of the Omnibus Crime Control and Safe Streets Act [22].

Despite these claims, the Clipper Chip faced overwhelming opposition. The most fundamental critique arose from its secrecy: the algorithm was classified. A cornerstone of modern cryptographic assurance is validation through public, adversarial testing. But by concealing the algorithm, the government forfeited an essential layer of trust, introducing fears

that Clipper Chip could house unintentional flaws or deliberate vulnerabilities. Without transparency, the public could not verify whether the system was secure or if it concealed weaknesses exploitable by malicious actors or even the government itself.

Furthermore, critics identified economic and geopolitical flaws in the initiative. While American companies would be compelled to adopt the Clipper Chip, foreign manufacturers faced no such constraints. This could cause consumers to turn to stronger, non-escrowed, foreign encryption alternatives [22]. This creates a landscape with weaker standards for US systems and unregulated access abroad.

Like Dual EC DRBG, Clipper Chip highlighted the inherent tensions between state surveillance goals and the principles underpinning secure communication systems. Both controversies illustrated how government efforts to institutionalize surveillance through cryptographic standards erode trust, not just in the specific technologies but in the broader institutions advocating their adoption. The resulting backlash – both domestic and international – fueled a growing push for independent and transparent standardization processes, culminating in efforts to reduce reliance on US-centric cryptographic ecosystems [22]. While proponents framed the chip as a measured compromise between privacy and security, its opacity, coercive implementation strategies, and flawed economic assumptions rendered it untenable [15]. In the end, the initiative's collapse underscored the necessity of transparent, inclusive processes in crafting policies that shape the foundations of global communication. Ultimately Clipper Chip remains a cautionary reminder that the costs of undermining trust often outweigh the purported benefits of control.

Another significant concern surrounding the Clipper Chip was the possibility of malicious third parties obtaining the escrowed keys. However, the Clipper Chip was only implemented in a single production phone, limiting its adoption and, with it, the incentive for third-party attacks [15]. In contrast, our next case study examines the CALEA Act, where a government backdoor was successfully exploited by malicious hackers.

The Calea Exploit

Similar to Clipper Chip, the Communications Assistance for Law Enforcement Act (CALEA) – passed in 1994 – mandates that telecom providers must make calls wiretappable and grant the government access to a customer's line if they obtain a legal order [16]. This act faced the same criticisms as Clipper Chip, namely that CALEA makes domestic hardware and software less attractive for global consumers and it would move research and development out of the US. However, unlike Clipper Chip, CALEA was successfully implemented, and wiretaps for all telecom providers in the US are in place to this day.

In 2024, a hacker group known as Salt Typhoon exploited the CALEA mandated wiretapping interfaces to infiltrate many major US telecom providers including Verizon, AT&T, and Lumen Technologies [23]. The hackers accessed call records and unencrypted text messages for thousands of Americans, including many senior government officials [24]. The full extent of the hack is unknown, and security researchers suspect that the group may still have access to US networks.

The ethical ramifications of this event are profound. The mandate for built-in surveillance capabilities introduced systemic vulnerabilities that were able to be exploited by malicious actors. This underscores another flaw with backdoors: even while intended to aid law

enforcement, they can compromise the privacy and security of all users. CALEA inadvertently compromised the very security it sought to protect, serving as a cautionary tale of how well-intentioned policies can lead to unintended consequences, compromising both individual privacy and national security.

Core Ethical Takeaways

The central ethical issue defining the debate over backdoors in cryptographic systems centers on balancing the imperatives of national security with the rights to privacy and trust in digital infrastructures. Though justified as necessary tools for combating crime and terrorism, their implementation often overlooks ethical and practical complexities.

First, backdoors are designed to selectively empower law enforcement while making the system weaker for everyone else. This duality makes backdoors uniquely vulnerable to abuse by unauthorized entities – as seen in the CALEA breach. The debate here does not center solely on trust, but is about whether introducing systemic vulnerabilities is a justified tradeoff when they can be equally weaponized against the very populations they are meant to protect.

Second, backdoors challenge the ethical legitimacy of governance. By mandating opaque systems, governments erode public confidence not only in specific technologies but in the institutions that promote them. The ethical question here is whether states, as custodians of security, can morally justify undermining the trust they rely on. Dual EC DRBG exemplifies how covert backdoors subvert the social contract between governments and citizens, which results in a broader crisis of confidence.

Third, backdoors can negatively reshape the economic and geopolitical landscape. Coercing domestic compliance incentivizes the decentralization of cryptographic innovation. This fracturing creates an uneven playing field where adversaries may have access to stronger cryptographic tools than are available domestically. This calls into question whether these policies are truly serving the public interest, or ultimately compromising national security.

Finally, backdoors raise questions about the limits of control and the ethical boundaries of surveillance. They compel us to ask if it is ethical to prioritize potential threats over present liberties. The assumption that backdoors will always be used responsibly, and only by “good” actors, is dangerous and has been demonstrably disproven multiple times. The ethical weight of technology lies in its use: once implemented, backdoors are neutral mechanisms, open to exploitation by any actor with sufficient motivation and intent. Going forward, an ethical design of cryptographic systems should prioritize minimizing harm over maximizing control.

Drawing from the case studies, we see that if backdoors are to exist, their development must adhere to principles of transparency, accountability, and minimal harm. Public scrutiny is a non-negotiable condition, as secrecy breeds mistrust. Moreover, independent oversight extending beyond state actors is critical to prevent misuse, recognizing that cryptographic systems underpin a complex global infrastructure. Ultimately, backdoors should strive to achieve both privacy and security wherever possible. The cases of Dual EC DRBG, Clipper Chip, and CALEA demonstrate the dangers of neglecting these principles, and highlight a need to reimagine the role of cryptography in society – not as a tool of control, but as a shared foundation for trust and security.

3. The Cypherpunk movement and Cybercrime

It is apparent that the government has a very fragile relationship with ethics and cryptography as we have noted in the previous sections. While the government must prioritize protecting the right to virtual privacy and anonymity for every user, it also must consider how to prevent crime and protect confidential government information without infringing on the prior. In this section, we will explore the responsibility cryptographers have to protect online freedoms, the connection between the government and cybercrime, and the ethical discussion of government and cryptography.

The Cypherpunk Manifesto

The Cypherpunk Manifesto, authored by Eric Hughes in 1993, is a foundational document of cryptographic ethics and privacy [29]. The manifesto highlights six fundamental responsibilities that lie on the shoulders of cryptographers who design, maintain, and sell cryptographic tools such as encryption and file storage [29]. These responsibilities include ensuring complete privacy for users, considering societal implications before releasing a tool, ensuring these devices are not used for unknown surveillance, preventing centralized control of a cryptographic system, finding the correct balance between protecting privacy and addressing security concerns, and anticipating new designs to match other technological standards even if the government does not specifically address cryptography [29].

Cryptographers work to ensure that their tools are made using the strongest known encryption algorithms, but it is essential that when users are interacting with these tools, there must be limited data collection of file content, user information, and purpose of usage [25]. Current standards are that data collection must be limited to the number of uses made by a user, size of files and information passed into cryptographic tools, network information to authenticate the user session, and temporary information about the plaintext that must be discarded after the session [25].

The reason why the information that is still recorded under top cryptography standards is related to storing tool usage information to further development designs and to address security concerns if needed [26]. Performance and numeric information about usage of cryptographic tools does not infringe on personal data privacy because confidentiality is maintained [26]. This information related to direct activity with an encryption tool can assist in criminal investigations where a suspect's information regarding encryption usage can solidify a case against them even if the actual user information cannot be recovered [26].

Wiretapping and surveillance with encrypted data is a topic cryptographers must adopt a strong stance against [30]. Government surveillance attempts such as the USA PATRIOT Act and TOLA Act give authorities the power to decrypt or access encrypted data in cases involving national security or criminal investigations [30]. These laws also provoke concerns about government overreach and the potential violation of privacy rights for innocent individuals [30]. Decryption laws can be tools for protecting national security, but they also create fears of surveillance and less civil liberties. Accessing encrypted data to prevent terrorism or crime might be seen as ethical to prevent harm. But it also involves privacy breaches that could have broader societal implications, especially if guilt is not certain for involved parties [30]. Cryptographers must have laws like these in mind when protecting private information of their

users and must make active efforts to curb laws that undermine the 4th amendment in cases of cryptography [30].

Cryptographers must recognize how their work impacts power structures and civil liberties [29]. As governments and corporations leverage cryptographic advancements for intelligence and surveillance, ethical dilemmas arise [25]. For instance, when encryption technologies heighten governmental control, they do so at the expense of individual freedoms [25]. The manifesto stresses decentralization as a safeguard against abuses of power, challenging the centralization tendencies of governments [25]. This theoretical framework underscores that while cryptography is a tool for security, it is also inherently political, influencing how authority and autonomy are distributed in society [25].

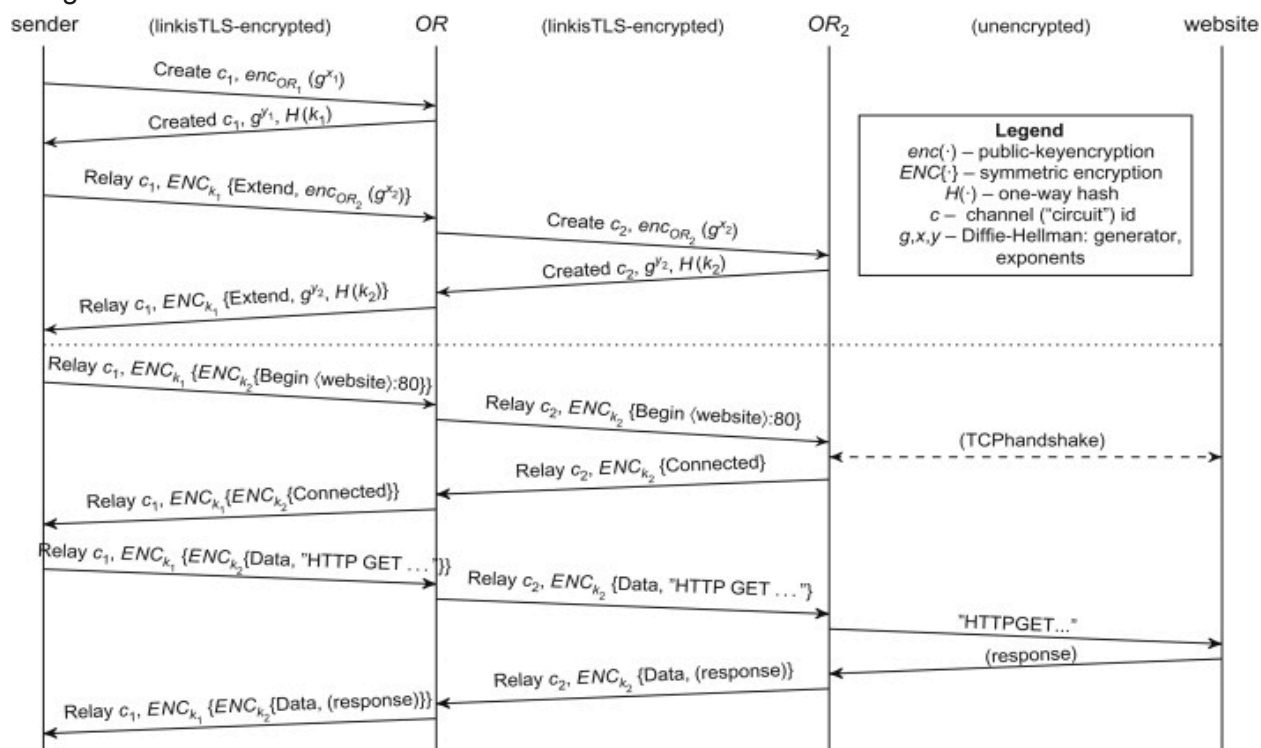
The manifesto's call for proactive cryptographic development aligns with the reality that legislation regulating technology often lags behind advancements [29]. Cryptographers are therefore tasked with ensuring that their innovations protect freedoms in rapidly evolving technological landscapes. By emphasizing ethical responsibility, the manifesto outlines a vision where cryptography acts as a force for decentralization, justice, and privacy—a vision as relevant today as it was in the early 1990s.

The Tor Browser

The Tor browser is an open-source platform managed by volunteer servers (Tor relays) across the world [31]. Much like Google Chrome or FireFox, Tor is a web browser that connects a user's host machine to a webpage on the internet. Tor supports different search engines, but the default is DuckDuckGo, an encrypted search engine that, much like Tor, does not store any user information such as session activity, IP address, and history [31]. Many people also use Tor to access services that regular browsers cannot reach, such as .onion sites which only function on the onion network (a network that only connects to web pages that are directed with onion routing) [28]. When you access a regular .com site through Tor the internet traffic is routed through a series of relays which hides your real IP address from the site you're visiting so the site only sees the IP of the Tor exit node [28]. The nearly complete anonymity of Tor and the DuckDuckGo search engine make the browser popular for cybercrime. A large example of this is the "Dark Web," a collection of networks and websites where the use of onion routing makes it nearly impossible to track user activity, allowing drug trafficking, stolen data markets, illegal arms deals, and more to operate without consequence [31].

Tor operates through onion routing, a secure encryption scheme that keeps a user completely anonymous to endpoint websites and eavesdroppers on networks that are known to the Tor browser [28]. Tor's architecture routes internet traffic through a series of relays, concealing the user's IP address and ensuring their activity cannot be easily traced [28]. The first step of onion routing is the creation of the circuit. When a user accesses the Tor network, the Tor client builds a random path (circuit) through multiple servers (nodes) in the Tor network [28]. The first step in the path is known as the entry node. This is the server that knows the user's IP address but does not know the destination of the user's request [28]. From this server, there are middle nodes which are servers that are randomly chosen to relay the traffic between the encrypted IP, endpoint, and request [28]. The exit node is the server that ends up with the endpoint address and the request the user sent, but it does not know the user's original IP address [28]. This is because a randomly chosen middle server decrypted the IP address and sent the request to another server that had the decrypted endpoint but not the original IP address [28].

While information is being passed between servers, the data is encrypted in layers, similar to an onion structure [28]. Each node in the circuit is a layer of encryption, and when the random circuit is built, the user's IP, request, and endpoint are encrypted n number of times where n is the number of nodes in the circuit [28]. Layers are peeled off sequentially as data traverses the circuit (decrypted), the exit node sends the request, keeping the original IP address anonymous, and responses from the destination site are re-encrypted by each node on the return path [28]. Additionally, no single node knows both the user's identity and destination, minimizing the risk of compromise [28]. The onion routing encryption scheme can be visualized in Figure 1 below:



[Figure 1] S. Fischer-Hbner, S. Berthold, *Computer and Information Security Handbook (Second Edition)*, 2013.

Tor was initially a project of the U.S. Naval Research Laboratory in the 1990s, and onion routing was designed to protect government communications by anonymizing their origin and destination [31]. Only government agents used .onion sites to convey confidential information, but an issue arose because adversaries were able to track this encrypted web activity and correlate it to government action because no other user had access to Tor or onion routing [31]. To make onion routing more effective in governmental use, the U.S. government authorized the release of Tor as an open-source tool [31]. From a cypherpunk manifesto stance, the release of Tor as a tool for the general public to utilize exemplifies the cypherpunk philosophy of government decentralization of encryption and allows more people to attain the maximum available privacy online [29]. However, the grave implication of releasing onion routing encryption to the public was the boom of cybercrime on anonymous Tor networks [31]. By democratizing access to anonymizing technology, the U.S. government enabled both legitimate privacy enhancements and the proliferation of illicit activities.

Some famous examples of the Tor browser being exploited for illegal activity are highlighted in the Silk Road Takedown and Operation Onymous led by the U.S. government

[27]. The Silk Road referred to a pioneering dark web marketplace for illegal drugs. The FBI traced Bitcoin transactions between suspected parties engaging on the site and were able to locate the servers that mainly ran the network [27]. Operation Onymous was an international government effort to take down over four hundred dark web sites suspected of trafficking, hired crime, and illegal selling of data. Authorities were able to exploit a vulnerability in the random generation algorithm of temporary Tor circuits used for onion routed browsing and eventually located Tor server locations to physically shut down illegal operations [27]. The vulnerability has since been closed by Tor volunteer servers [27].

The Ethical Dilemma

Cryptographers working individually or in the government face many ethical obligations when introducing cryptographic tools to the general public. Specifically addressing the Tor browser, cryptographers working for the government authorized the release of Tor and the onion routing encryption scheme providing enhanced privacy for users. This aligns with the goal of enhancing user privacy in the Cyberpunk Manifesto. However, this also led to the creation of the dark web. Cryptographers must consider whether the benefits of cryptographic tools outweigh the harm they could cause if misused by malicious parties [32].

Tor's ethical complexities are emphasized in cases where the anonymity it provides conflicts with law enforcement objectives. For instance, there was a case where the U.S. government shut down an illegal pornography site running on a Tor supported server [32]. However, the government was unable to successfully prosecute the parties involved in the crime because revealing investigative methods would compromise onion routing's security, suggesting that the government has some method of breaking Tor encryption [32]. While the revelation of new tools like this decryption would jeopardize privacy of those who use Tor for legal purposes, going against the Cypherpunk Manifesto, it is also unjust to conceal new cryptographic tools from the public at the expense of the justice system [32]. This raises crucial considerations for cryptographers to consider when dealing with cryptographic tools. Should government cryptographers prioritize dismantling criminal networks or protect the online privacy of both the general public and the government [32]. Additionally, if the government has cryptographic power and tools that are not known to or available to the public, how much of that power should the government exercise to ensure that the rights to privacy and security of data online are protected [32].

The Cypherpunk Manifesto's principles remain as a guide in the age of Tor and advanced cryptographic systems. As governments and cryptographers navigate the tension between privacy and security, they must consider how their actions align with ethics and following the existing constitution of rights. By addressing ethical dilemmas proactively, cryptographers can ensure that their cryptographic advancements serve as tools for empowerment rather than oppression.

4. Freedom of Expression

In the 1996 paper "Cryptography and the First Amendment: The Right to be Unheard," Phillip Reiman spoke about the positive and negative impacts of cryptography as it relates to maintaining the first amendment. In this, cryptography is likened to the printing press, as it

enables us to have a brand new and revolutionary way to spread information with each other. In response to this advancement, the National Security Agency's General Council declared in 1994 that "citizens do not have a constitutional right to unbreakable encryption algorithms." Around this time, two arguments arose for the regulation of cryptography: 'cryptography is not speech, and therefore not subject to the first amendment' and 'cryptography is speech, but the rules of free speech do not apply to cryptography[33].' Throughout the course of this paper, Reiman argues that (necessarily strong) cryptography is vital for the freedom of Americans, and it must be upheld and included in the protections provided by the first amendment of our constitution[33].

From the view of someone now living in 2024, it can be seen that this stance on cryptography reflects how common and open-source cryptography has become in modern life. With most people carrying smartphones, we are all constantly relying on cryptographic principles to keep our information both safe from tampering/accidental change and private. While it is unlikely anyone would argue the importance of cryptography in our society, the complete spread of it has also meant that some agents and governments have used it to actually decrease the free speech of certain groups of internet users.

The spread of cryptographic ideas has given rise to a new field of study and law in our society: cybersecurity. When the general public thinks of cybersecurity, they think of experts who are writing complex programs or hackers breaking into a mainframe, but we use cybersecurity principles every day when we make a new password for a website or log into our phones. Some governments have taken advantage of this concept of cybersecurity and safety, and use it as an excuse to take away freedoms from their citizens[34].

In 2007, Saudi Arabia put into effect the Anti-Cyber Crime Law. While positive sounding in concept, if one looks further into the fine print, it can be seen that the law is intended to infringe on the freedom of speech of the country's citizens. To begin, one of the stated goals of this law was "protecting public interest, morals, and common values[35]." This leaves the government to be able to state what qualifies as against the public interest or common values, and prosecute accordingly. This is further continued when "the production, preparation, transmissions, or storage of material impinging on public order, religious values, public morals, or privacy, through an information network or computer[35]" is stated to be a crime punishable by up to five years in prison. While it is difficult to find exact cases where this law has been used against citizens attempting to speak up against injustices, it can easily be seen how this could be used by those in power to silence those who disapprove of any governmental practices.

Any person who commits any of the following cybercrimes shall be subject to imprisonment for a period not exceeding five years and a fine not exceeding ١٠,٠٠٠,٠٠٠ riyals, or either penalty:

- l. The production, preparation, transmission, or storage of material impinging on public order, religious values, public morals, or privacy, through an information network or computer.
- ٢. The construction or publication of a web site on an information network or computer to promote or facilitate human trafficking.
- ٣. The preparation, publication, or promotion of material for pornographic networks or gambling activities which violates public morals.
- ٤. The construction or publication of a web site on an information network or computer to trade, distribute, demonstrate methods of use, or facilitate dealing in narcotic and psychotropic drugs.

The sentence may provide for the publication of its summary at the expense of the convicted person in one or more local newspapers or in any other appropriate medium, depending on the type, gravity, and impact of the crime committed, provided that such publication is executed after the judgment becomes final.

As someone living in the United States, one may see this and dismiss it as something that only happens in 'oppressive' countries far away, but movements such as this one have been attempted in Britain. In 2015, British Prime Minister David Cameron expressed an interest in banning the use of any encrypted messaging applications if British intelligence was not given access through things such as backdoors or complete access[36]. This was a part of a movement that took place around European countries after the terrorist attack in Paris. Mr. Cameron was on record for the New York Times saying: "Are we going to allow a means of communications which it simply isn't possible to read? No, we must not[36]."

While this law was never carried out, movements such as this should give the general public alarm, as they demonstrate a clear desire for the governments of 'free' countries to break the cryptographic protections that we have built ourselves.

According to Reiman, the best way to prevent this centralization of all cryptography and encrypted communication is to maintain the free market, and maintain the individual's right and ability to encrypt their own communication in a way that is not readable by any power[33]. If the cryptographic were to be overtaken by a single monopoly, this company/organization would be extremely vulnerable to both 3rd party hacks that break the entire system and governmental pressures/lobbying that allow for backdoors into the personal information of every user of their service.

To prevent things such as this, we must stay vigilant and watch out for any movements in our society toward a world in which people no longer have the right to control and regulate their own cryptography or any laws which side with companies that are becoming monopolies in their areas, even if this field is unrelated to cryptography.

References

- [1] J. F. Dooley, History of Cryptography and Cryptanalysis: Codes, Ciphers, and Their Algorithms. Cham: Springer International Publishing, 2018.
- [2] W. Diffie, "Post-Quantum Cryptography in Context," William Gould Dow Distinguished Lecture, University of Michigan, Nov. 2024.
- [3] G. Krasniqi, "Importance of Cryptography in the Government," in 2021 UBT International Conference, A. Pefqeli, Ed., University for Business and Technology in Kosovo, Nov. 2021.
- [4] B. Buchanan, "Cryptography and Sovereignty," Survival, vol. 58, no. 5, pp. 95–122, Sep. 2016, doi: <https://doi.org/10.1080/00396338.2016.1231534>.
- [5] S. Miller and T. Bossomaier, "Privacy, Encryption and Counter-Terrorism," Advanced Sciences and Technologies for Security Applications, pp. 139–154, 2021, doi: https://doi.org/10.1007/978-3-030-90221-6_9.
- [6] D. Moore and T. Rid, "Cryptopolitik and the Darknet," Survival, vol. 58, no. 1, pp. 7–38, Jan. 2016, doi: <https://doi.org/10.1080/00396338.2016.1142085>.

- [7] B. Gellman and A. Soltani, "NSA infiltrates links to Yahoo, Google data centers worldwide, Snowden documents say," Washington Post, Oct. 30, 2013. Available: https://www.washingtonpost.com/world/national-security/nsa-infiltrates-links-to-yahoo-google-data-centers-worldwide-snowden-documents-say/2013/10/30/e51d661e-4166-11e3-8b74-d89d714ca4dd_story.html
- [8] J. Borger, J. Ball, and G. Greenwald, "Revealed: how US and UK spy agencies defeat internet privacy and security," the Guardian, Sep. 06, 2013. <https://www.theguardian.com/world/2013/sep/05/nsa-gchq-encryption-codes-security>
- [9] H. Abelson et al., "Keys under doormats: mandating insecurity by requiring government access to all data and communications," Journal of Cybersecurity, vol. 1, no. 1, pp. 69–79, Sep. 2015, doi: <https://doi.org/10.1093/cybsec/tyv009>.
- [10] J.D. Tuccille, "Chinese Hackers Used U.S. Government-Mandated Wiretap Systems," Reason.com, Oct. 11, 2024. URL: <https://reason.com/2024/10/11/chinese-hackers-used-u-s-government-mandated-wiretap-systems/>
- [11] U. H. R. C. (47th sess : 2021 : Geneva), "The promotion, protection and enjoyment of human rights on the Internet :: resolution /: adopted by the Human Rights Council on 13 July 2021," digitallibrary.un.org, Jul. 2021, Available: <https://digitallibrary.un.org/record/3937534?ln=en&v=pdf>
- [12] NIST, "Federal Information Processing Standards Publication 197 Announcing the ADVANCED ENCRYPTION STANDARD (AES)," 2001. Available: <https://nvlpubs.nist.gov/nistpubs/FIPS/NIST.FIPS.197.pdf>
- [13] Jennifer A. Martin. Encryption Backdoors: A Discussion of Feasibility, Ethics, and the Future of Cryptography. Digital Commons at Seattle Pacific University. 2017. URL: <https://digitalcommons.spu.edu/honorsprojects/69>.
- [14] Daniel J. Bernstein, Tanja Lange, and Ruben Niederhagen. Dual EC: A Standardized Back Door. Cryptology ePrint Archive, Paper 2015/767. 2015. URL: <https://eprint.iacr.org/2015/767>.
- [15] Vandana Pednekar-Magal and Peter Shields. The State and Telecom Surveillance Policy: The Clipper Chip Initiative. Communication Law and Policy. 2010. URL: https://doi.org/10.1207/S15326926CLP0804_03.
- [16] Emily Hancock. "9. CALEA: Does One Size Still Fit All?" In: Digital Cops in a Networked Environment. Ed. by Jack Balkin et al. New York, USA: New York University Press, 2007, pp. 184–204. ISBN: 9780814739334. DOI: 10.18574/nyu/9780814739334.003.0011. URL: <https://doi.org/10.18574/nyu/9780814739334.003.0011>.

[17] K. Marton et al. "Generation and testing of random numbers for cryptographic applications". In: Proceedings of the Romanian Academy - Series A: Mathematics, Physics, Technical Sciences, Information Science. 13 (Oct. 2012), pp. 368–377.

[18] Dan Shumow and Niels Ferguson. On the Possibility of a Back Door in the NIST SP800-90 Dual Ec Prng. CRYPTO 2006. 2006. URL: <https://rump2007.cr.yp.to/15-shumow.pdf>.

[19] Joseph Menn. "Exclusive: Secret contract tied NSA and security industry pioneer". In: Reuters (Dec. 30, 2017). URL: <https://www.reuters.com/article/world/exclusive-secret-contract-tied-nsa-and-security-industry-pioneer-idUSBRE9BJ1C5/> (visited on 12/12/2024).

[20] Michael Rogers and Grace Eden. "The Snowden Disclosures, Technical Standards, and the Making of Surveillance Infrastructures". In: International Journal of Communication 11 (2017), pp. 802–823.

[21] Nadiya Kostyuk and Susan Landau. "Dueling Over Dual_ec_drbg: TheConsequences of Corrupting a Cryptographic Standardization Process. In: Harvard National Security Journal 13 (2022), pp. 224–284.

[22] Howard S. Dakoff. "The Clipper Chip Proposal: Deciphering the Unfounded Fears That Are Wrongfully Derailing Its Implementation". In: UIC Law Review 29 (1996), pp. 475–498.

[23] Maggie Miller. "Chinese hack of global telecom providers is 'ongoing,' officials warn". In: Politico (Dec. 3, 2024). URL: <https://www.politico.com/news/2024/12/03/chinese-hack-global-telecom-ongoing-00192410> (visited on 12/12/2024).

[24] Maggie Miller. "Krouse, Volz, Viswanatha, and McMillan". In: The Wall Street Journal (Oct. 5, 2024). URL: <https://www.wsj.com/tech/cybersecurity/u-s-wiretap-systems-targeted-in-china-linked-hack-327fc63b> (visited on 12/12/2024).

[25] E. Blair, "Cryptography at the Crossroads: Ethical Responsibility, the Cyberpunk Movement and Institutions," June. 2024, <https://eprint.iacr.org/2024/970.pdf>.

[26] G. Thomas, "Discussing Privacy and Ethical Issues of Encryption in a Connected World," Oct. 2020, https://auscomputerethics.com/wp-content/uploads/2021/03/aice_2020_paper_8.pdf.

[27] E. Jardine, A. Lindner, G. Owenson "The potential harms of the Tor anonymity network cluster disproportionately in free countries," Nov. 2020, <https://pmc.ncbi.nlm.nih.gov/articles/PMC7749358/>.

- [28] S. Fischer-Hbner, S. Berthold, *Computer and Information Security Handbook (Second Edition)*, 2013.
- [29] Eric Hughes. A cypherpunk's manifesto, 1993.
- [30] W. Diffie and S. Landau. Privacy on the Line: The Politics of Wiretapping and Encryption. MIT Press, 2001.
- [31] P. Story*, D. Smullen, R. Chen, Y. Yao, A. Acquisti, L. Faith Cranor, N. Sadeh*, F. Schaub, "Increasing Adoption of Tor Browser Using Informational and Planning Nudges," sciendo, 2022. <https://petsymposium.org/popets/2022/popets-2022-0040.pdf>.
- [32] T. Collison, "The Ethical Considerations of the Tor Browser," Medium, Jan. 2015. <https://medium.com/applied-cryptography/the-ethical-considerations-of-the-tor-browser-8bbc415720b6>.
- [33] Reiman, P. E. (1996). *Cryptography and the First Amendment: The right to be unheard*, 14 J. Marshall J. Computer & Info. L. 325 (1996). UIC Law Open Access Repository. <https://repository.law.uic.edu/jitpl/vol14/iss2/6/>
- [34] Rossi, C., & Green, N. (n.d.). Cybersecurity and human rights. Public Knowledge. <https://www.gp-digital.org/wp-content/uploads/2015/06/GCCS2015-Webinar-Series-Introductory-Text.pdf>
- [35] *Anti-cyber crime law*. تفاصيل النظام. (n.d.). <https://laws.boe.gov.sa/BoeLaws/Laws/LawDetails/25df73d6-0f49-4dc5-b010-a9a700f2ec1d/2>
- [36] Frizell, S. (2015, January 13). *British prime minister floats ban on apps like Snapchat and Whatsapp*. Time. <https://time.com/3665270/david-cameron-snapchat-whatsapp-terror/>